Final Term Project Report

Name: Muhammad Affan Siddiqui

Student Number: 260418005

Lesson Name: Visual Programming

Lesson Number: COM206

INTRODUCTION

Project Title:

UniTrack — Smart Lost & Found Management System

Research Problem / Problem Statement:

Universities frequently face difficulties in managing lost and found items within campuses. In traditional systems, when an item is found by students or university staff, it is usually submitted manually to the administration office. Staff members often attach handwritten tags containing location and date information, making the tracking process inefficient and disorganized.

These manual methods create several problems:

- Difficulty in tracking lost items
- Unauthorized claims of items
- Lack of centralized records
- Poor communication among staff members
- Time-consuming management process

Additionally, returned and unclaimed items are difficult to monitor properly, resulting in confusion and loss of records.

This project addresses the problem by developing a desktop-based Lost & Found Management System using PyQt and SQLite database technologies. The system digitizes item registration, searching, claim verification, and returned item tracking through an interactive graphical user interface.

Objective of the Project:

The objective of this project is to develop a smart desktop application that helps university administration efficiently manage lost and found items.

The system allows administrators to:

- Add found items into the system

- Store records digitally using SQLite database
- Search items dynamically
- Track active and returned items
- Verify claims securely
- Manage lost item information through a user-friendly graphical interface

The project also demonstrates visual programming concepts including:

- GUI development using PyQt5
- Event-driven programming
- Database integration
- Table management
- User interaction handling
- Responsive interface design

Expected Outcome:

The expected outcome of this project is a fully functional desktop application capable of simplifying the lost and found management process inside universities.

The system enables users to:

- Log into the system securely
- Add and store found item information
- Search active lost items dynamically
- Mark items as returned
- Track statistics of active and returned items
- Display data using interactive tables
- Improve overall item management efficiency

Overall, the project demonstrates how visual programming technologies can solve real-world administrative problems through automation and digital record management.

METHODOLOGY

Approach to Problem Solving:

1. Control Flow of the System:

The system follows an event-driven and database-driven architecture where user interactions trigger backend processes dynamically.

When administrators interact with features such as login, item registration, item searching, or claim verification, PyQt widgets process events and interact with the SQLite database accordingly.

2. Input, Processing, Output:

Input:

- Admin login credentials
- Item name
- Category
- Found location
- Item description
- Search keywords
- Claim verification actions

Processing:

- PyQt processes user events dynamically
- SQLite database stores and retrieves records
- Search queries filter item data
- Claim verification updates item status
- Dashboard statistics update automatically
- GUI components refresh dynamically

Output:

- Display of active lost items
- Dashboard statistics
- Search results table
- Returned item tracking
- Dynamic graphical user interface updates
- Confirmation and validation messages

3. System Control Flow Steps:

1. User opens UniTrack application
2. Login window appears
3. Admin enters credentials
4. Dashboard screen opens
5. Admin adds found item records
6. Data is stored in SQLite database
7. Search system retrieves active items
8. Returned items are verified and updated
9. Dashboard statistics refresh dynamically
10. Users continue managing lost and found operations

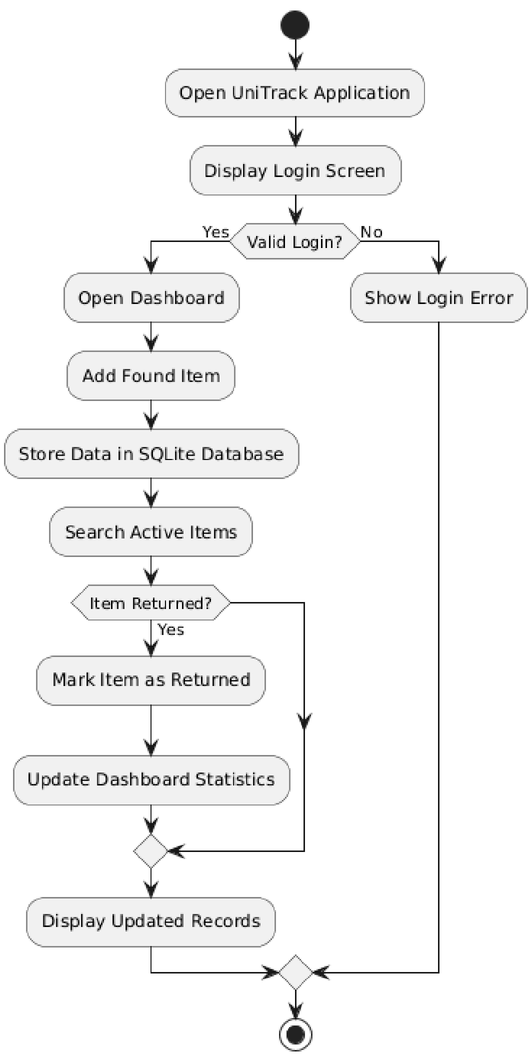

Figure 1: System Flowchart of UniTrack

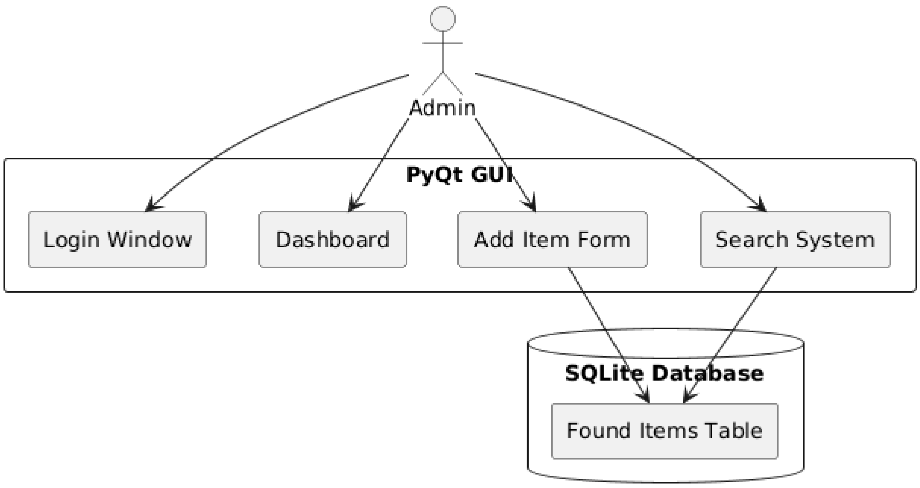

Figure 2: System Architecture and Component Interaction

TECHNICAL IMPLEMENTATION

Programming Languages:

- Python
- PyQt5
- SQLite Database
- Qt Widgets
- Object-Oriented Programming
- Event-Driven Programming

Code Example:

```python
def save_item(self):

    item_name = self.item_input.text()
    category = self.category_input.text()

    connection = sqlite3.connect("unitrack.db")
    cursor = connection.cursor()

    cursor.execute("""
    INSERT INTO found_items
    (item_name, category)

    VALUES (?, ?)
    """, (item_name, category))

    connection.commit()
    connection.close()
```

Explanation:

The project is implemented using PyQt5 framework for graphical user interface development and SQLite for database management.

The system follows an event-driven programming approach where user actions trigger backend processing dynamically.

The application contains multiple windows including:

- Login Window
- Dashboard Window
- Add Item Window
- Search Window

SQLite database integration enables permanent storage of item records and returned item tracking.

The dashboard dynamically updates statistics for:

- Active lost items
- Returned items

Additionally, table widgets provide interactive searching and record display functionality.

The project demonstrates practical implementation of:

- GUI development
- Database integration
- Object-oriented programming
- Event handling
- Dynamic table management
- Real-time statistics updates

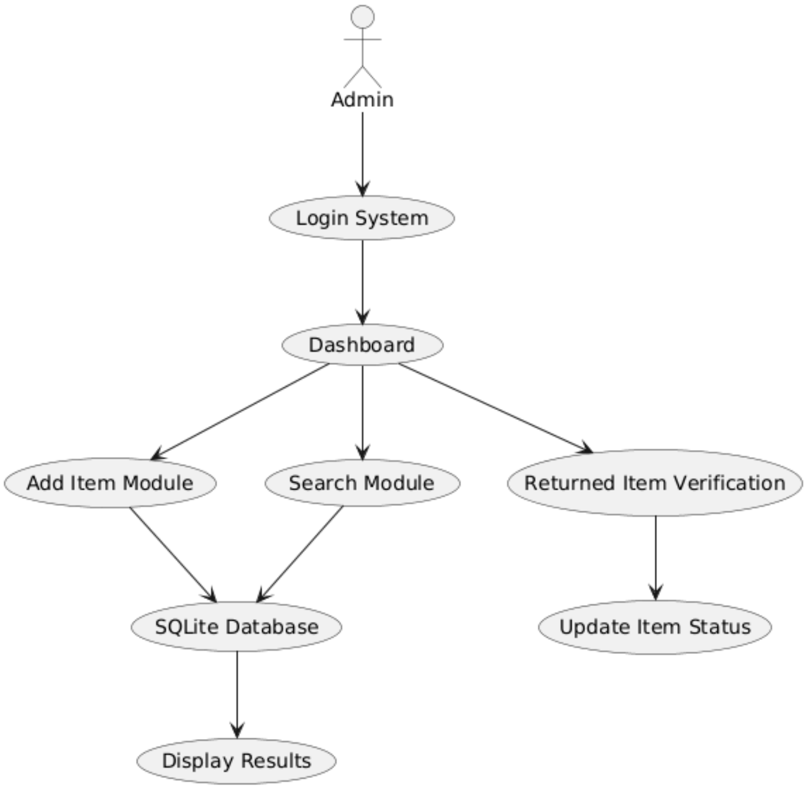

Figure 3: Data Flow

Conclusion:

UniTrack successfully demonstrates how visual programming technologies can be used to solve real-world administrative challenges faced by universities.

The system provides a centralized and efficient solution for managing lost and found operations through a modern desktop application.

The project highlights important software engineering and visual programming concepts such as:

- GUI development using PyQt5
- Event-driven programming
- SQLite database management
- Dynamic searching
- Interactive table systems
- Claim verification mechanisms

Overall, the project provides both technical implementation experience and practical social impact by improving lost item tracking and management efficiency within university environments.